

600V Half-Bridge Driver

PRODUCT SUMMARY

- V_{OFFSET} 600 V max.
- $I_{\text{O}+/-}$ (min) 130 mA / 270 mA
- V_{OUT} 10 V - 20 V
- $t_{\text{on/off}}$ (typ.) 100 ns / 130 ns
- **Dead Time (typ.)** 520 ns

GENERAL DESCRIPTION

The HPD2606X is a high voltage, high speed power MOSFET and IGBT drivers. Proprietary HVIC and latch immune CMOS technologies enable ruggedized circuit construction. The logic input is compatible with standard CMOS or LSTTL output, down to 3.3 V logic. The output drivers feature a high pulse current buffer stage designed for minimum driver cross conduction. The floating channel can be used to drive an N-channel power MOSFET or IGBT in the high-side configuration which operates up to 600 V.

FEATURES

- Floating channel designed for bootstrap operation
- Fully operational to +600 V
- Tolerant to negative transient voltage, dV/dt immune
- Gate drive supply range from 10 V to 20 V
- Under voltage lock out
- 3.3 V, 5 V and 15V logic compatible
- Cross-conduction prevention logic
- Matched propagation delay for both channels
- Outputs in phase with inputs
- Embedded dead time
- RoHS compliant
- SOP8 package

TYPICAL APPLICATION CIRCUIT

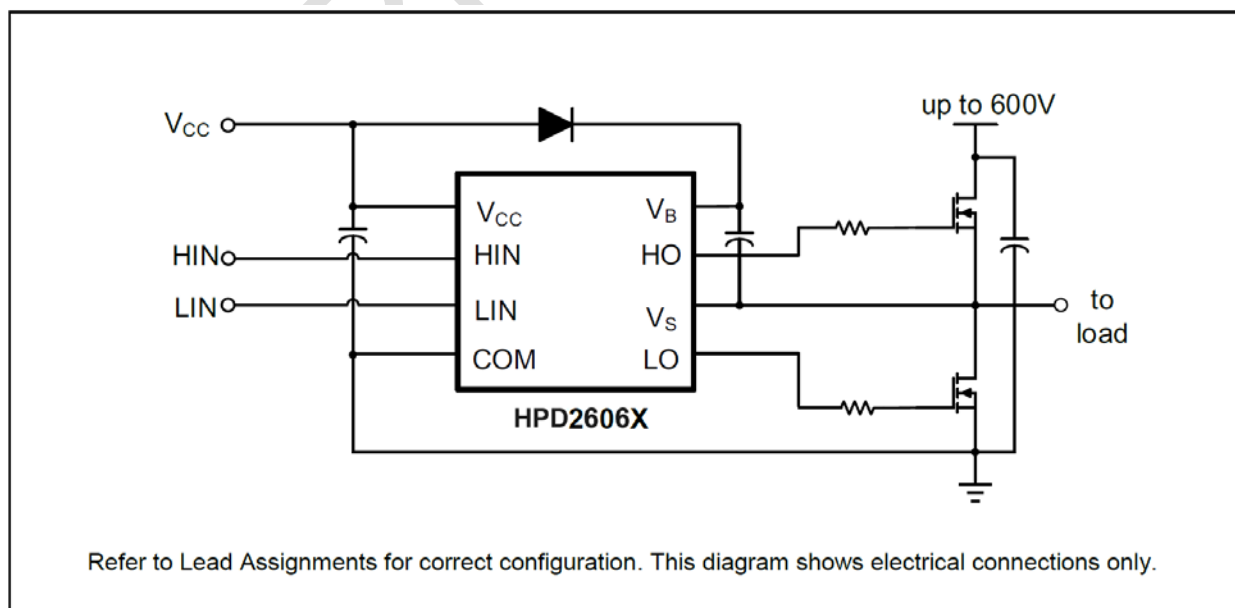


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1. PIN CONFIGURATION

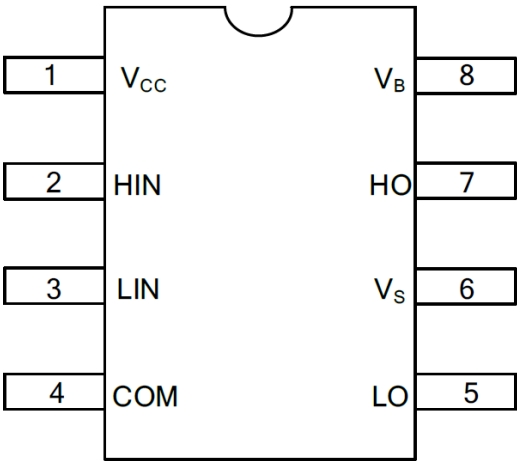
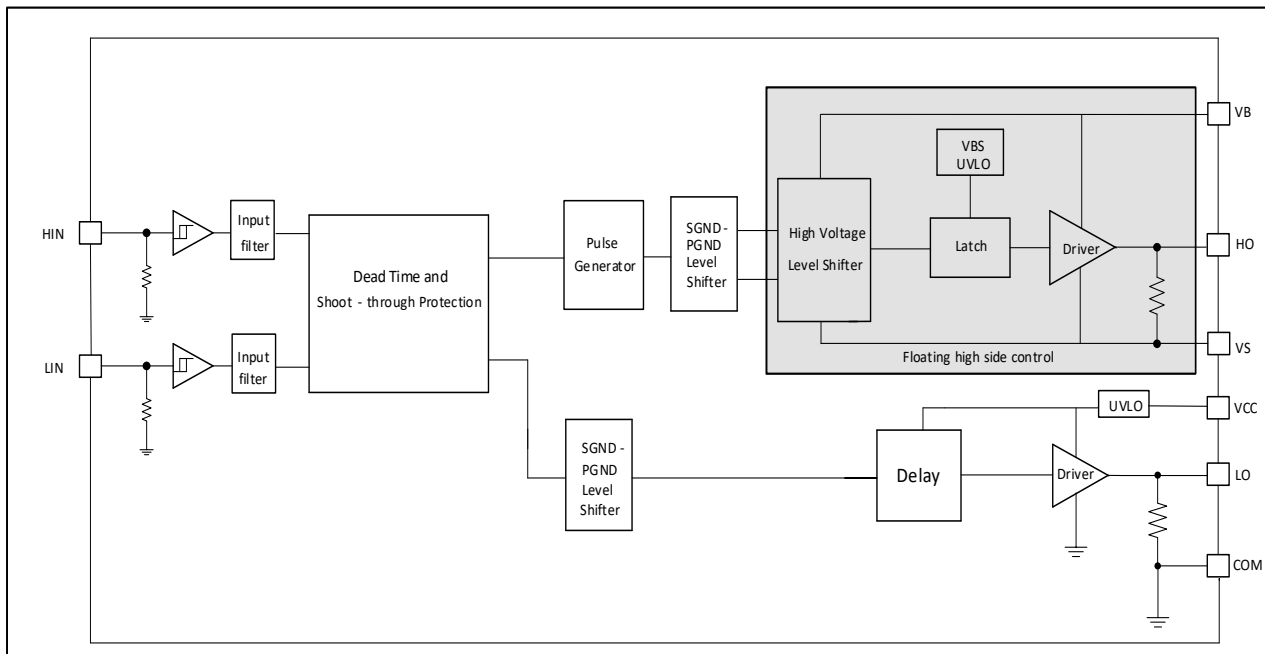


Figure 1-1. HPD2606X SOP Pin Arrangement

Table 1-1. HPD2606X Pin Description

No.	Pin	Description
1	V_{CC}	Low-side and logic fixed supply
2	HIN	Logic input for high-side gate driver output (HO), in phase
3	LIN	Logic input for low-side gate driver output (LO), in phase
4	COM	Low-side return
5	LO	Low-side gate drive output
6	V_S	High-side floating supply return
7	HO	High-side gate drive output
8	V_B	High-side floating supply

2. FUNCTIONAL BLOCK DIAGRAM



3. ABSOLUTE MAXIMUM RATINGS

Symbol	Definition	Min.	Max.	Units
V _B	High-side floating absolute voltage	-0.3	625	V
V _S	High-side floating supply offset voltage	V _B - 25	V _B + 0.3	
V _{HO}	High-side floating output voltage	V _S - 0.3	V _B + 0.3	
V _{CC}	Low-side and logic fixed supply voltage	-0.3	25	
V _{LO}	Low-side output voltage	-0.3	V _{CC} + 0.3	
V _{IN}	Logic input voltage (HIN & LIN)	-0.3	V _{CC} + 0.3	
dV _S /dt	Allowable offset supply voltage transient	---	50	V/ns
P _D	Package power dissipation @ T _A ≤ +25°C	---	0.625	W
θ _{JA}	Thermal resistance, junction to ambient	---	200	°C/W
T _J	Junction temperature	---	150	°C
T _S	Storage temperature	-55	150	

Note: Absolute maximum ratings indicate sustained limits beyond which damage to the device may occur. All voltage parameters are absolute voltages referenced to COM. The thermal resistance and power dissipation ratings are measured under board mounted and still air conditions.

4. RECOMMENDED OPERATION CONDITIONS

Symbol	Definition	Min.	Max.	Units
V _B	High-side floating absolute voltage	V _S + 10	V _S + 20	V
V _S	High-side floating supply offset voltage		600	
V _{HO}	High-side floating output voltage	V _S	V _B	
V _{CC}	Low-side and logic fixed supply voltage	10	20	
V _{LO}	Low-side output voltage	0	V _{CC}	
V _{IN}	Logic input voltage (HIN & LIN)	0	V _{CC}	
T _A	Ambient temperature	- 40	125	°C

Note: For proper operation the device should be used within the recommended conditions. The V_S offset rating is tested with all supplies biased at a 15 V differential.

5. DYNAMIC ELECTRICAL CHARACTERISTICS

V_{BIAS} (V_{CC} , V_{BS}) = 15 V, C_L = 1000 pF and T_A = 25°C unless otherwise specified.

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
t_{on}	Turn-on propagation delay	$V_S = 0$ V	---	100	150	ns
t_{off}	Turn-off propagation delay	$V_S = 0$ V	---	130	180	
t_r	Turn-on rise time		---	70	170	
t_f	Turn-off fall time		---	25	90	
DT	Deadtime, LS turn-off to HS turn-on & HS turn-on to LS turn-off		400	520	650	

6. STATIC ELECTRICAL CHARACTERISTICS

V_{BIAS} (V_{CC} , V_{BS}) = 15 V and T_A = 25°C unless otherwise specified. The V_{IN} , V_{TH} , and I_{IN} parameters are referenced to COM and are applicable to all logic input leads: HIN and LIN. The V_O and I_O parameters are referenced to COM and are applicable to the respective output leads: HO or LO.

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
V_{IH}	Logic "1" input voltage	$V_{CC} = 10$ V to 20V	2.5	---	---	V
V_{IL}	Logic "0" input voltage		---	---	0.8	
V_{OH}	High level output voltage, $V_{BIAS} - V_O$	$I_O = 2$ mA	---	0.05	0.2	
V_{OL}	Low level output voltage, V_O		---	0.02	0.1	
I_{LK}	Offset supply leakage current	$V_B = V_S = 600$ V	---	---	50	μ A
I_{QBS}	Quiescent V_{BS} supply current	$V_{IN} = 0$ V	---	50	65	
I_{QCC}	Quiescent V_{CC} supply current		---	120	150	
I_{IN+}	Logic "1" input bias current	$V_{IN} = 5$ V	---	8	15	
I_{IN-}	Logic "0" input bias current	$V_{IN} = 0$ V	---	---	5	
V_{CCUV+} V_{BSUV+}	V_{CC} & V_{BS} supply undervoltage positive going threshold		8	8.8	9.6	V
V_{CCUV-} V_{BSUV-}	V_{CC} & V_{BS} supply undervoltage negative going threshold		7.4	8.15	9	
I_{O+}	Output high short circuit pulsed current	$V_O = 15$ V, $V_{IN} =$ Logic "1", $PW \leq 10$ μ s	130	290		mA
I_{O-}	Output low short circuit pulsed current	$V_O = 0$ V, $V_{IN} =$ Logic "0", $PW \leq 10$ μ s	270	600		

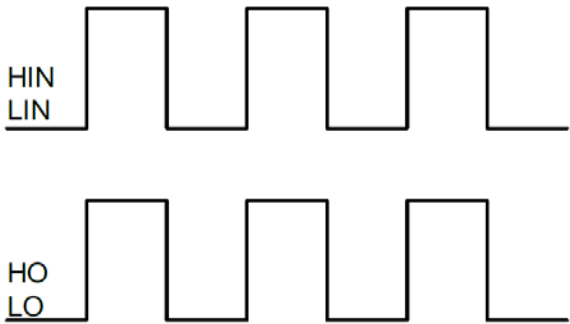


Figure 6-1. Input/Output Timing Diagram

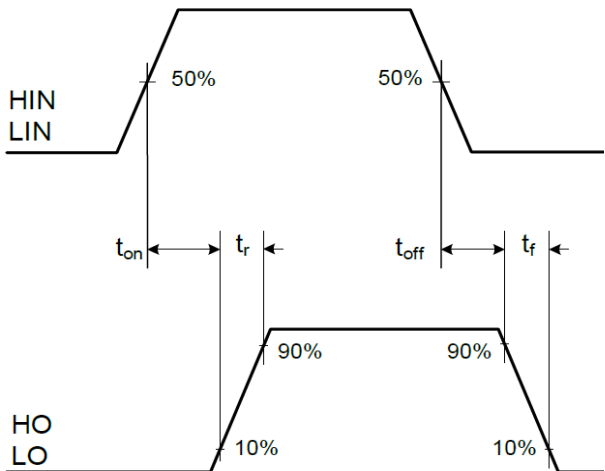


Figure 6-2. Switching Time Waveform

7. TYPICAL CHARACTERISTIC

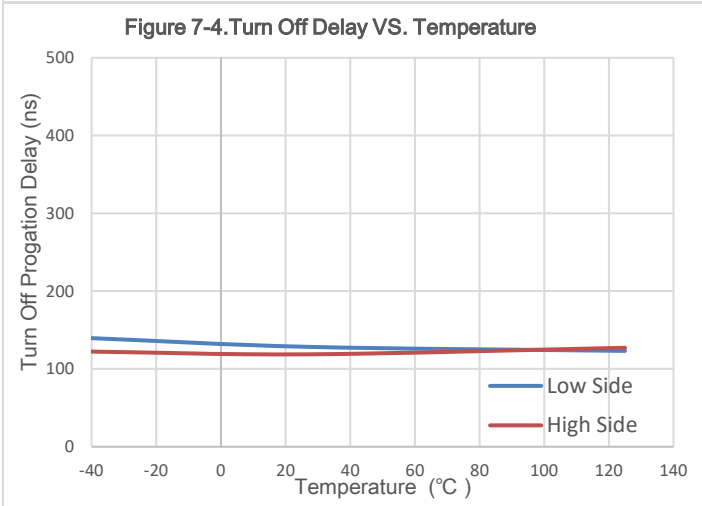
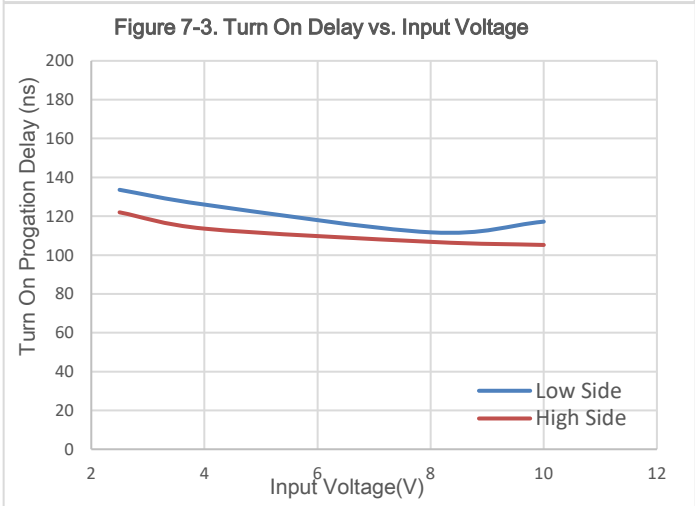
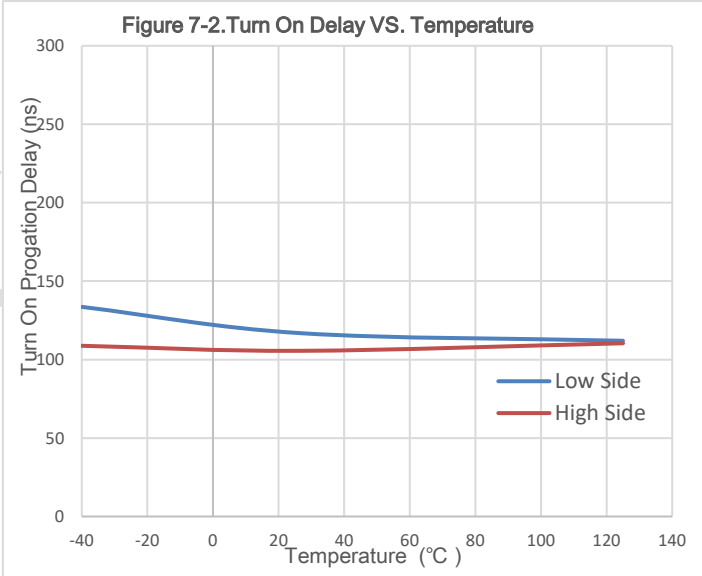
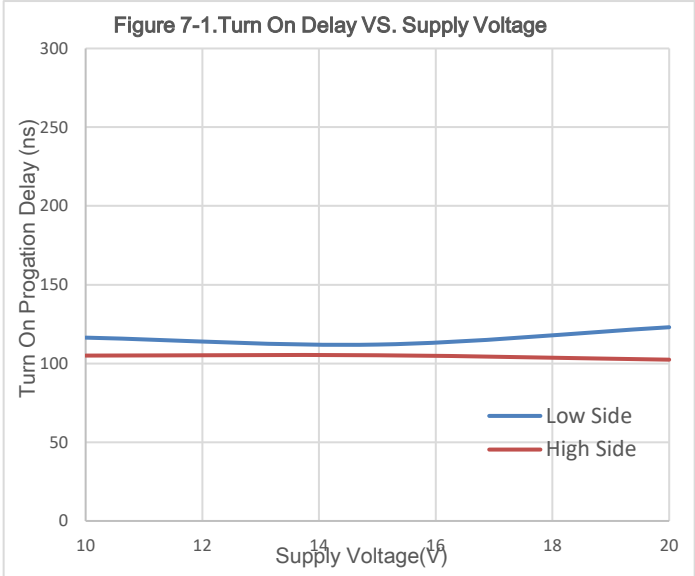


Figure 7-5. Turn Off Delay vs. Supply Voltage

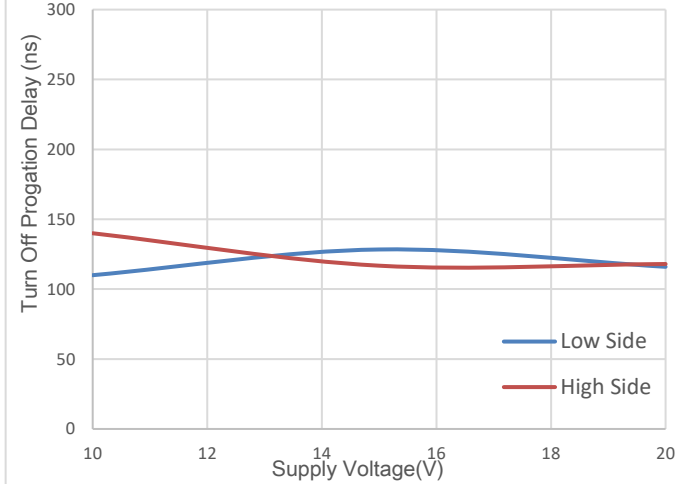


Figure 7-6. Turn Off Delay vs. Input Voltage

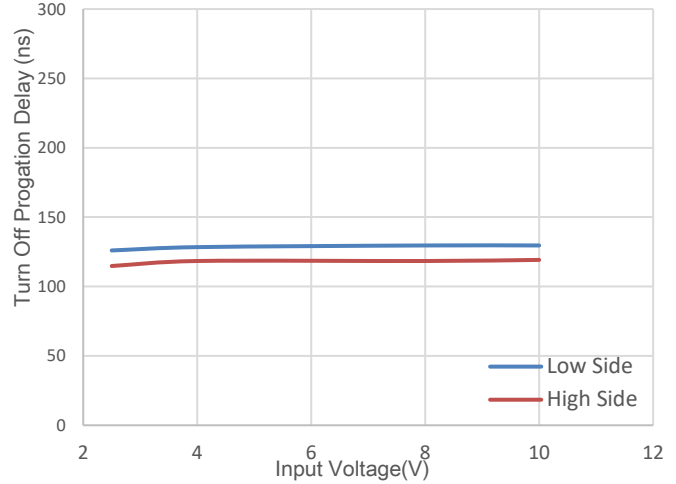


Figure 7-7. Turn On Rise Time VS. Supply Voltage

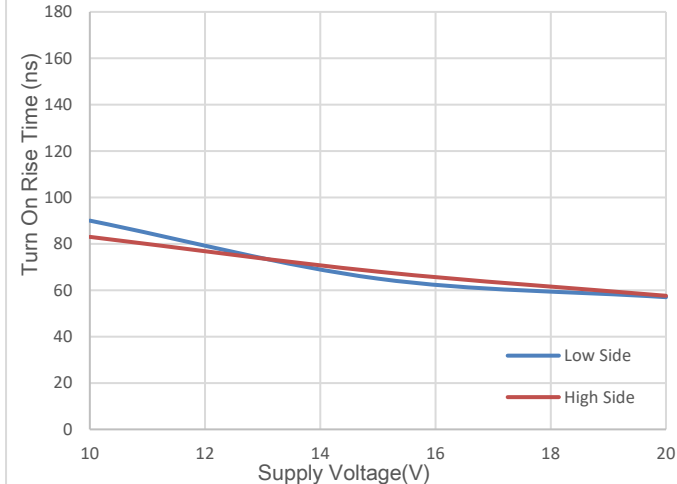


Figure 7-8. Turn On Rise Time VS. Temperature

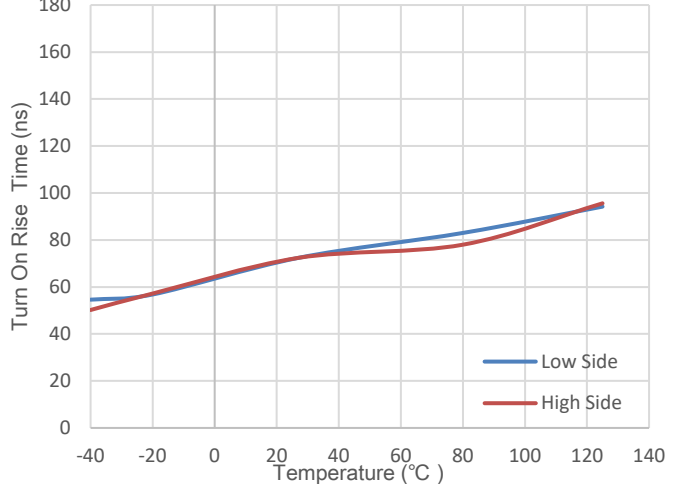


Figure 7-9. Turn Off Fall Time VS. Temperature

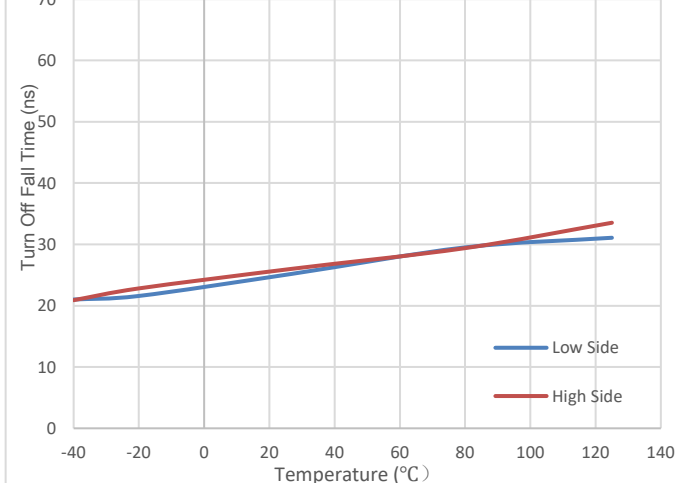


Figure 7-10. Turn Off Fall Time VS. Supply Voltage

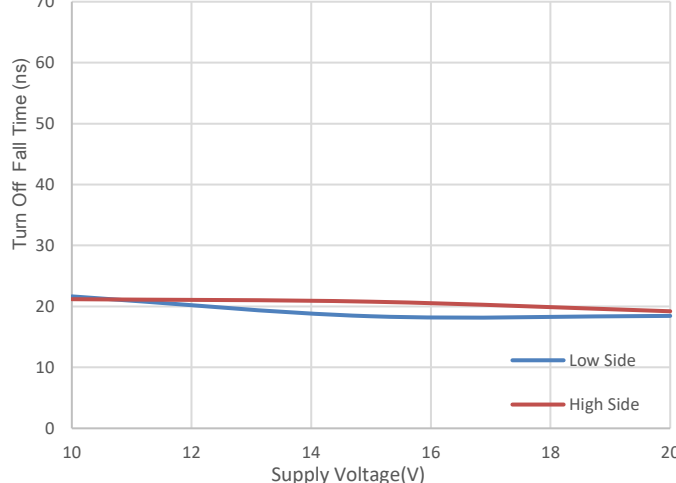


Figure 7-11. VB=600V Offset Supply Current VS.

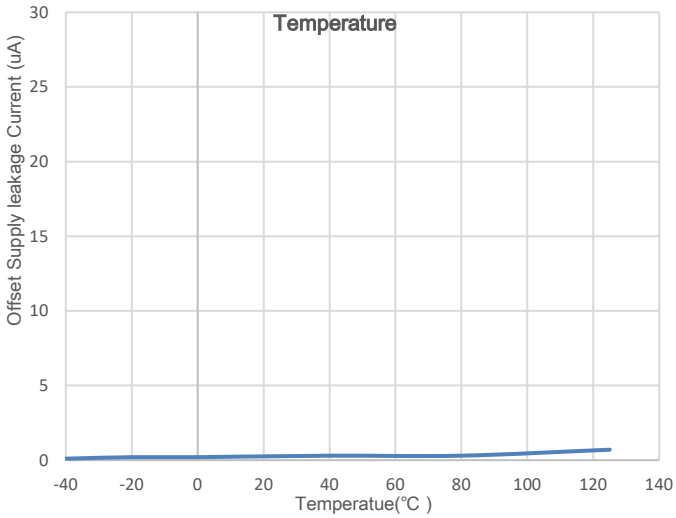


Figure 7-12. VB=600V Offset Supply Current vs. VB

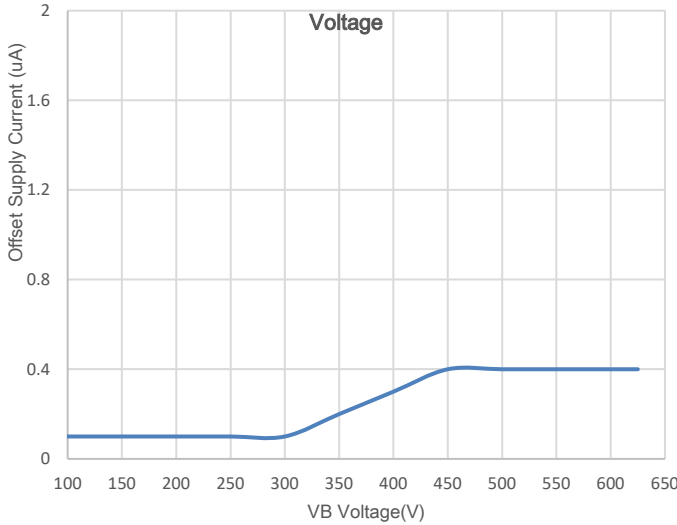


Figure 7-13. DT VS. Temperature

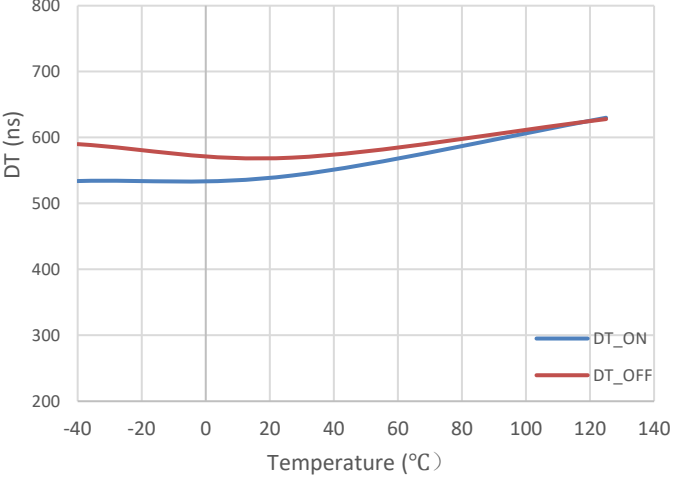


Figure 7-14. DT VS. Supply Voltage

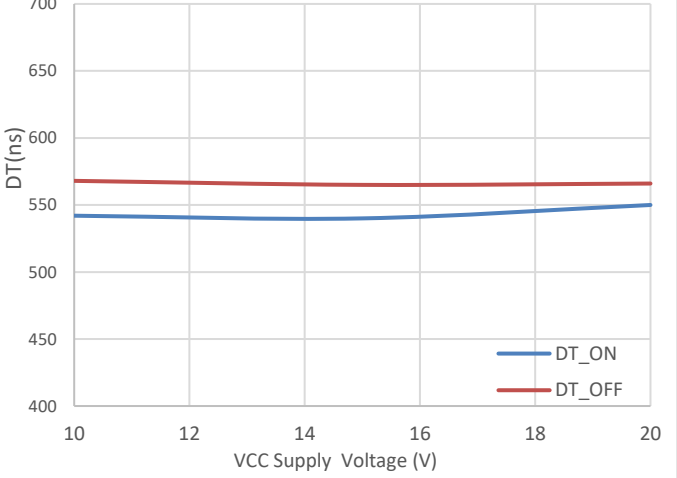


Figure 7-15. High Level Output VS. Temperature

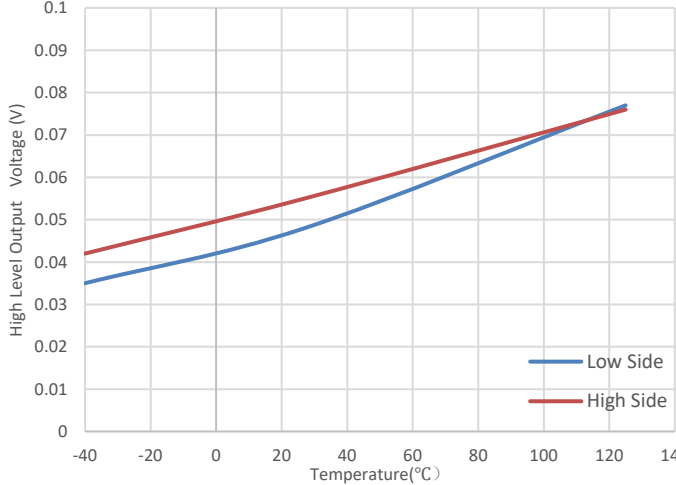


Figure 7-16. High Level Output VS. Supply Voltage

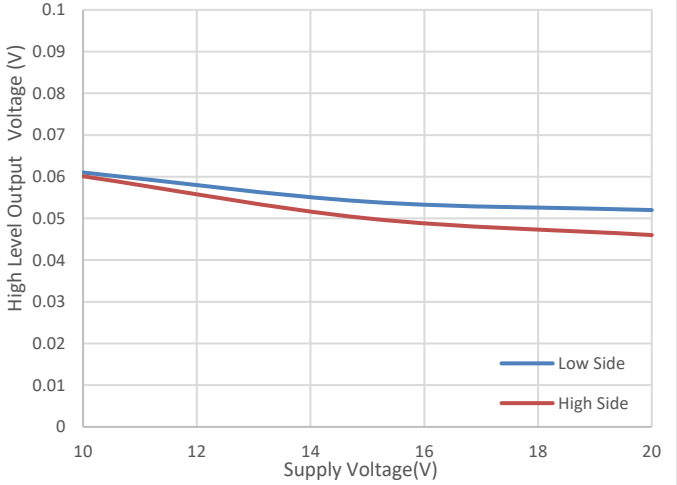


Figure 7-17. Low Level Output VS. Temperature

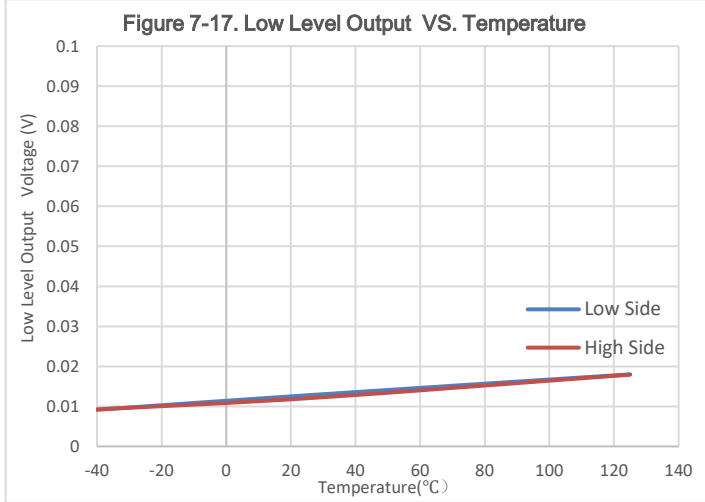


Figure 7-18. Low Level Output VS. Supply Voltage

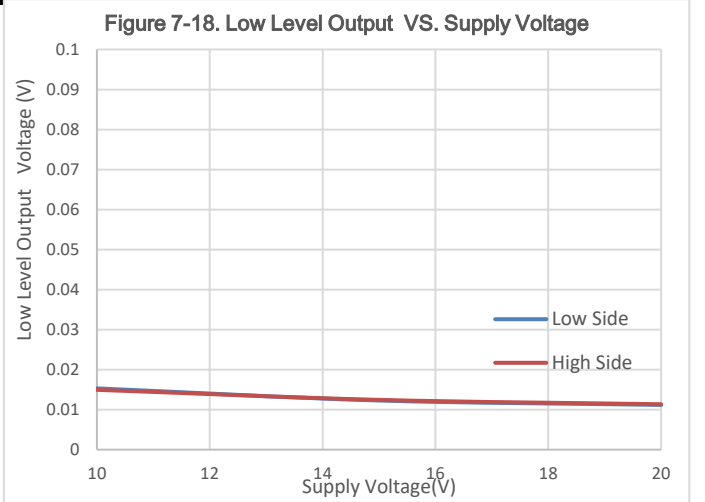


Figure 7-19. Offset Supply Current VS. Temperature

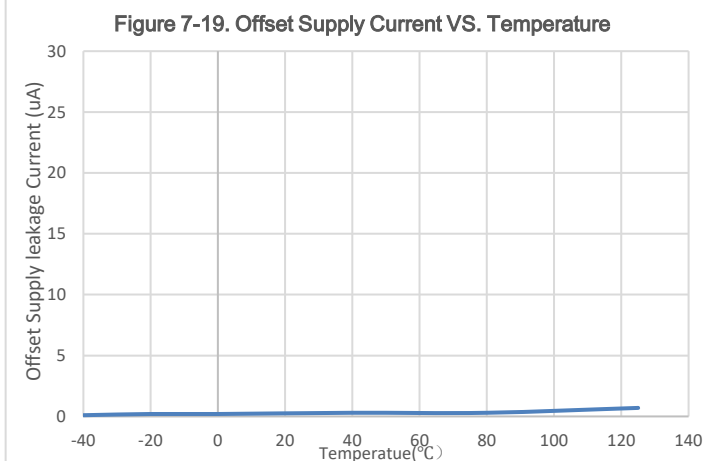


Figure 7-20. Offset Supply Current VS. VB Voltage

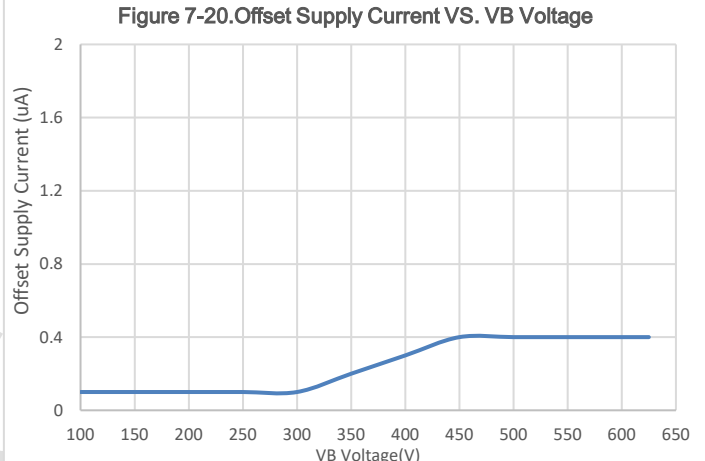


Figure 7-21 .VBS Supply Current VS. Temperature

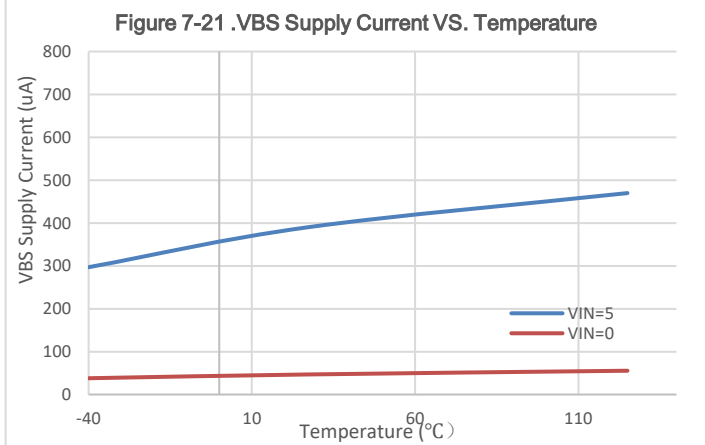
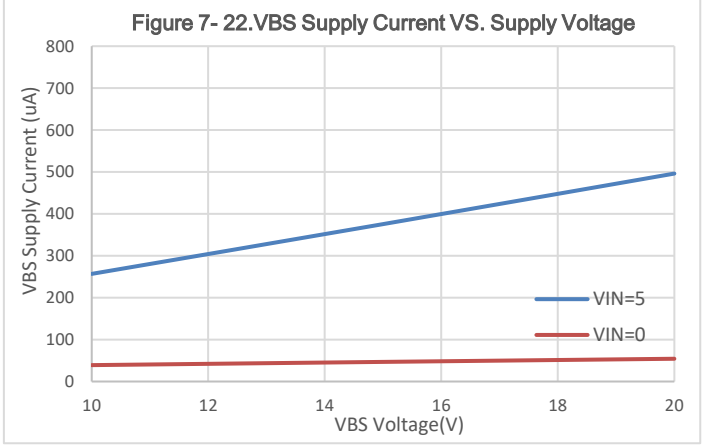
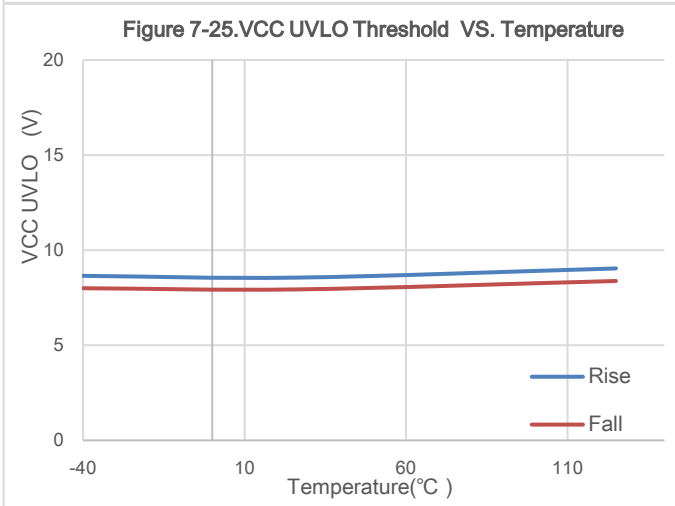
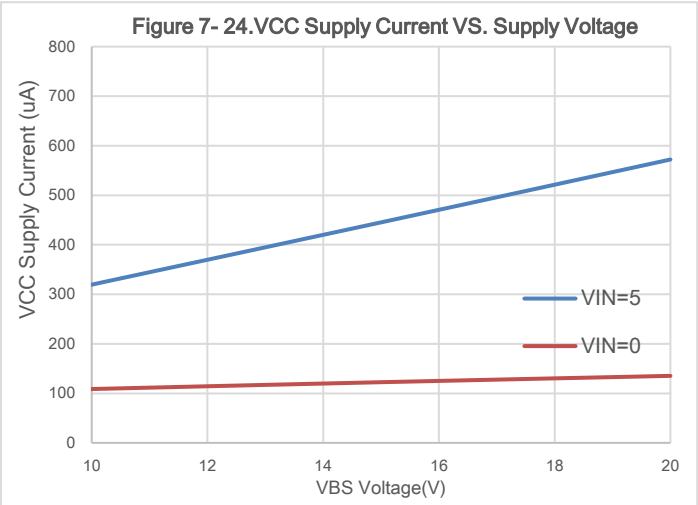
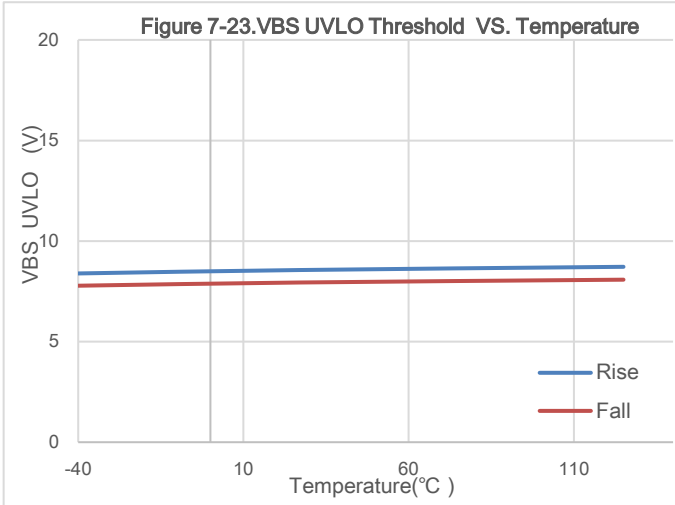


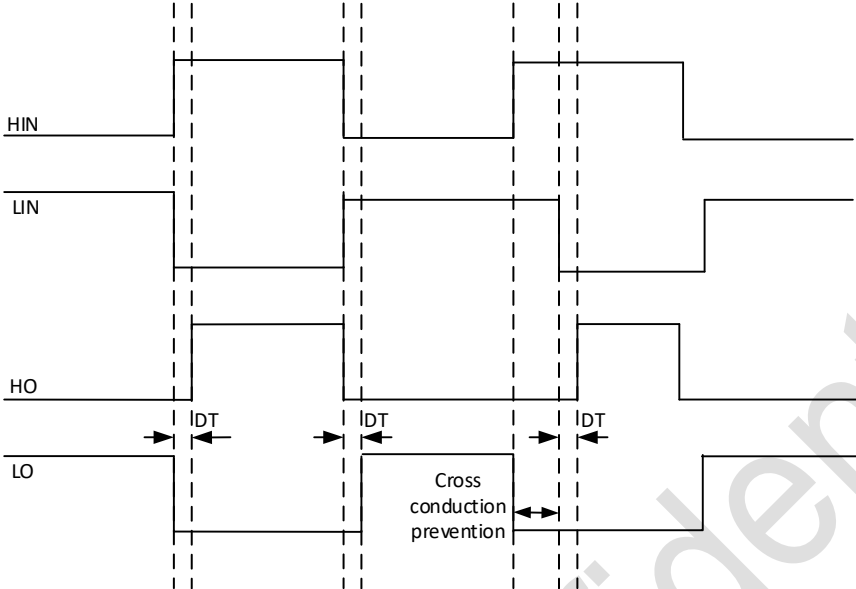
Figure 7-22.VBS Supply Current VS. Supply Voltage





8. CROSS-CONDUCTION PREVENTION

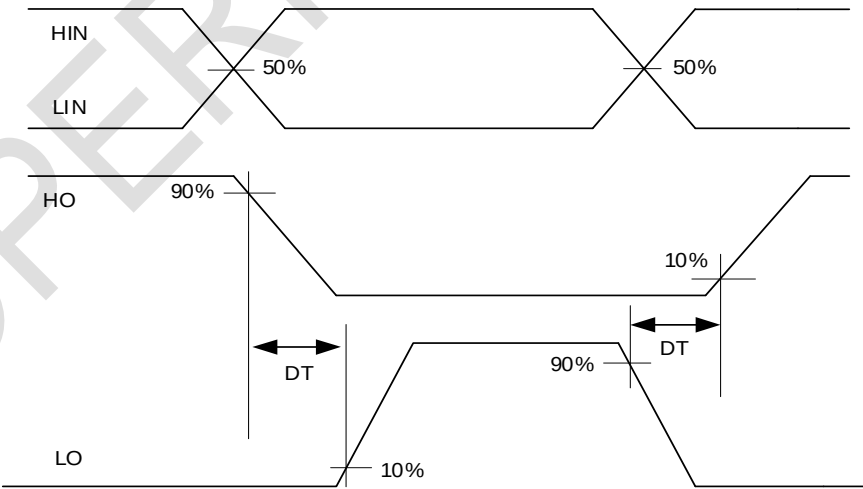
The embeded protection circuit designed for corss conduction prevention can effectively save the power tube from being damaged when the high side and low side input signals are interfered.
The following diagram shows how the prevention circuit works.



9. DEAD TIME FUNCTION

A fixed dead-time protection circuit is set up inside the chip. At dead time, both the high side and low side outputs are set to low. In order to effectively prevent the cross conduction of the upper and lower tubes, the set dead time must ensure that one power tube is turned off before opening another power tube. If the external dead time set by the logic input is longer than the internal dead time, the external dead time set by the logic input is configured as the output dead time. If the external dead time set for the logic input is shorter than the internal dead time, the output dead time is configured as the internal dead time.

The following diagram shows the timing relationship between dead time, input signal, and output signal.



10. ORDER INFORMATION

INDUSTRIAL RANGE: -40°C to +125°C

Order Part No.	Package	QTY
HPD2606X	SOP8	1000/Reel

11. PACKAGE OUTLINE

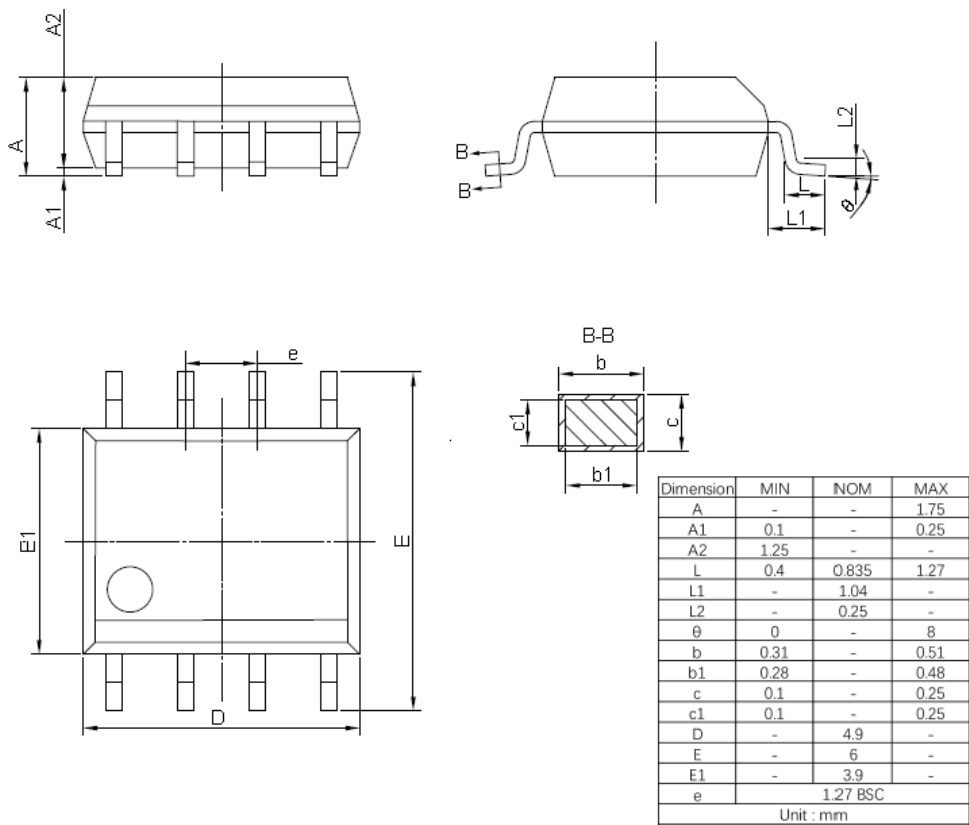


Figure 11-1. SOP8 Outline Dimensions

12. CONTACTS

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